Review

Hydrogenic atom

1-*e* Schrödinger equation

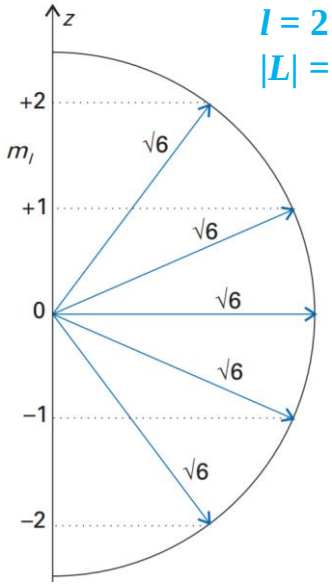
$$\left(\frac{-\hbar^2}{2\mu}\nabla^2 - \frac{Ze^2}{4\pi\varepsilon_0 r}\right)\psi(r,\theta,\phi) = E\psi(r,\theta,\phi)$$

$$\psi(r,\theta,\phi) = R(r)Y(\theta,\phi)$$

$$\Lambda^2 Y = -l(l+1)Y \qquad \textbf{Angular part}$$

$$-\frac{\hbar^2}{2\mu}\frac{\mathrm{d}^2 u}{\mathrm{d}r^2} + V_{\text{eff}}u = Eu \qquad u = rR \qquad \textbf{Radial part}$$

$$E_n = -\frac{Z^2\mu e^4}{32\pi^2\varepsilon_0^2\hbar^2 n^2}$$



***l* = 2**
$|L| = \sqrt{6}\hbar$

Principal quantum number *n*, determines the energy, ***n* = 1, 2, 3, ...**

Orbital angular momentum quantum number *l*

$|L| = \{l(l+1)\}^{1/2}\hbar$ **$l = 0, 1, 2, \ldots, n-1$**

Magnetic quantum number *m*

a *z*-component of angular momentum **$m\hbar$, $m = 0, \pm 1, \pm 2, \ldots, \pm l$**

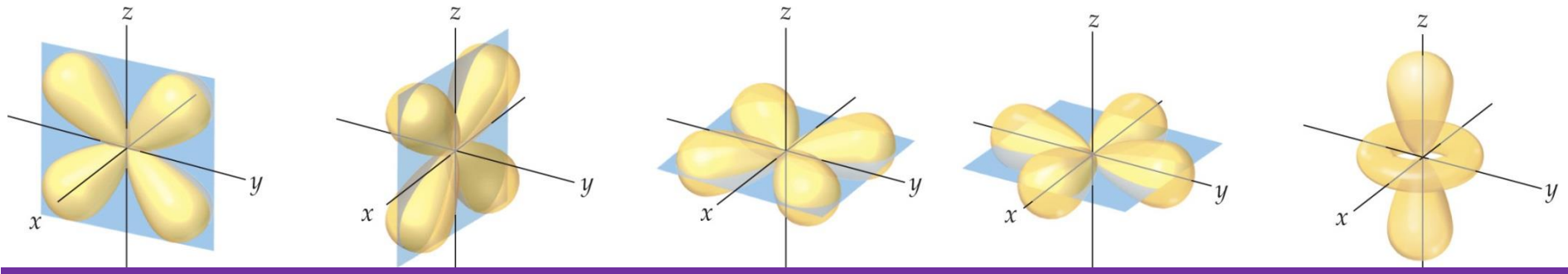
Review

An atomic orbital: a one-electron wavefunction for an electron in an atom, defined by **the 3 quantum numbers, *n*, *l*, *m*.**

Specify a state of an electron, *n*, *l*, m_l, m_s

The energy levels

$$E_n = -\left(\frac{Z^2 \mu e^4}{32\pi^2 \varepsilon_0^2 \hbar^2}\right)\frac{1}{n^2} \qquad n = 1, 2, \ldots$$



1 2 3 4 5

Review

Structures of many-electron atoms

Orbital Approximation: think of each electron as occupying its 'own' orbital.

$$\psi(r_1, r_2, \ldots) = \psi(r_1)\psi(r_2)\ldots$$

Electronic Configuration: the list of occupied orbitals.

Penetration and shielding

Building-up principle

For a ground state:

a. **Pauli exclusion principle**: Every orbital may contain only two electrons of opposite spins.

b. **The principle of minimum energy**: Whilst being compatible with the Pauli principle, electrons occupy the orbital with the lowest energy first.

c. **Hund's rule**: In degenerate energy states, electrons tend to occupy as many degenerate orbitals as possible. (Adopt a configuration with the greatest number of unpaired electrons.)

1*s* 2*s* 2*p* 3*s* 3*p* 4*s* 3*d* 4*p* 5*s* 4*d* 5*p* 6*s*

Review

Atomic terms $^{2S+1}L_J$

$(2L + 1)(2S + 1)$ terms

Selection rules for spectral transition *in light atoms*

$\Delta S=0$; $\Delta L=0,\pm1$; $\Delta l=\pm1$;

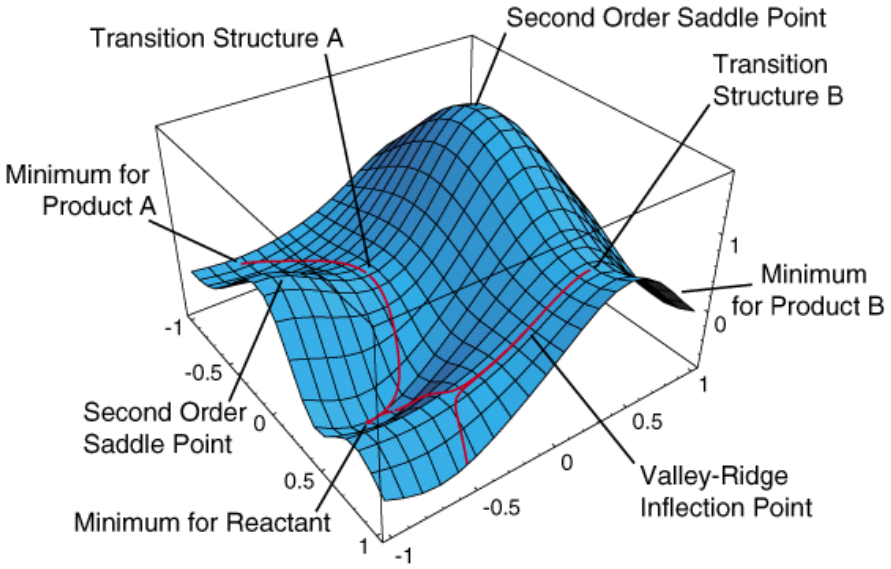
$\Delta J=0,\pm1$, but $J=0$ to $J=0$ is forbidden

In heavy atoms when *jj*-coupling is appropriate,

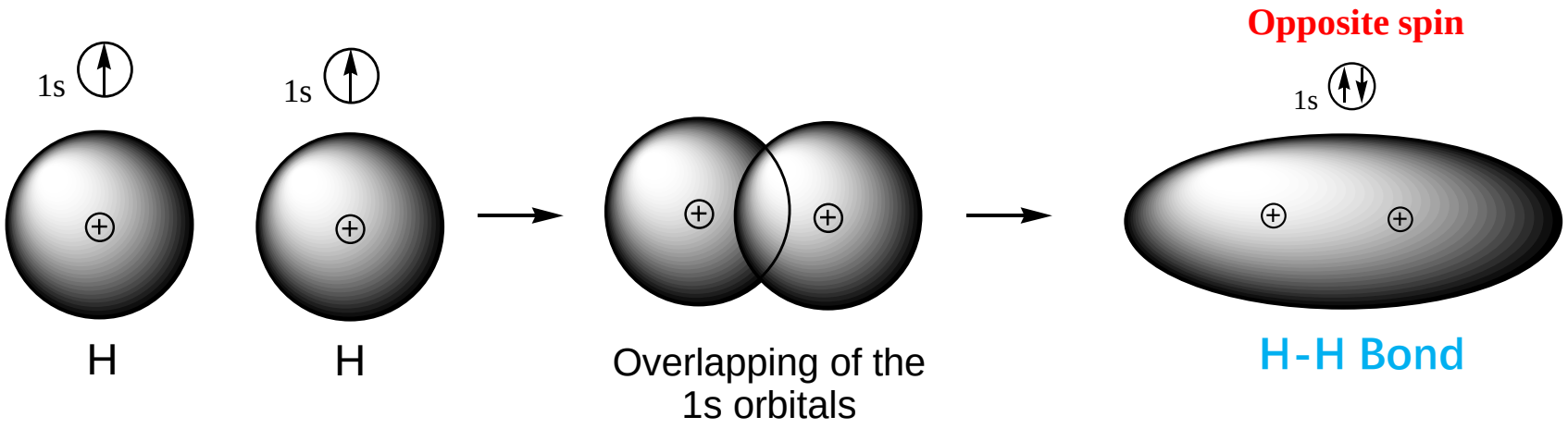
$\Delta S=\pm1$ are allowed.

The Born-Oppenheimer Approximation

→ **Solve the Schrödinger equation of electrons alone**



Valence-Bond Theory (价键理论)



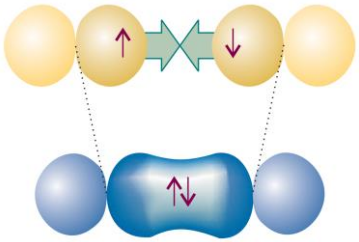
A bond is regarded as forming when an electron in an atomic orbital on one atom pairs its spin with that of an electron in an atomic orbital on another atom.

Covalent bond: one in which the spins of the two electrons pair as the atomic orbitals overlap.

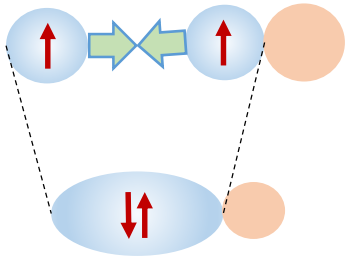
σ, π bonds



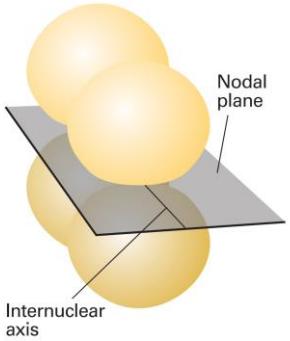
$s + s$



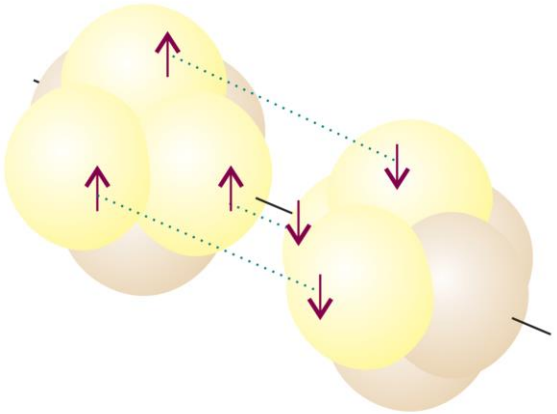
$p_z + p_z$



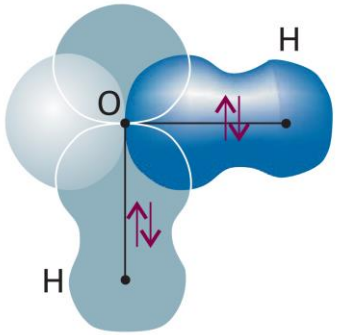
σ bond



π bond

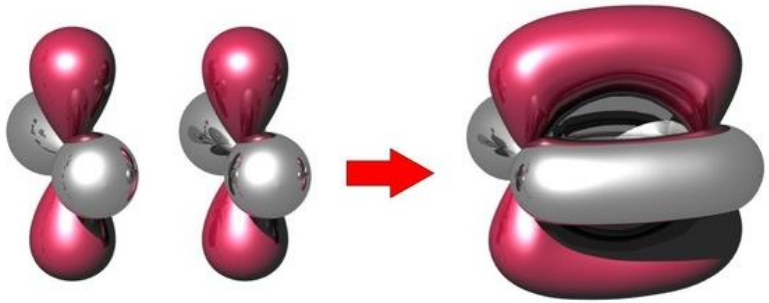


N_2: $\sigma + 2\pi$ bonds



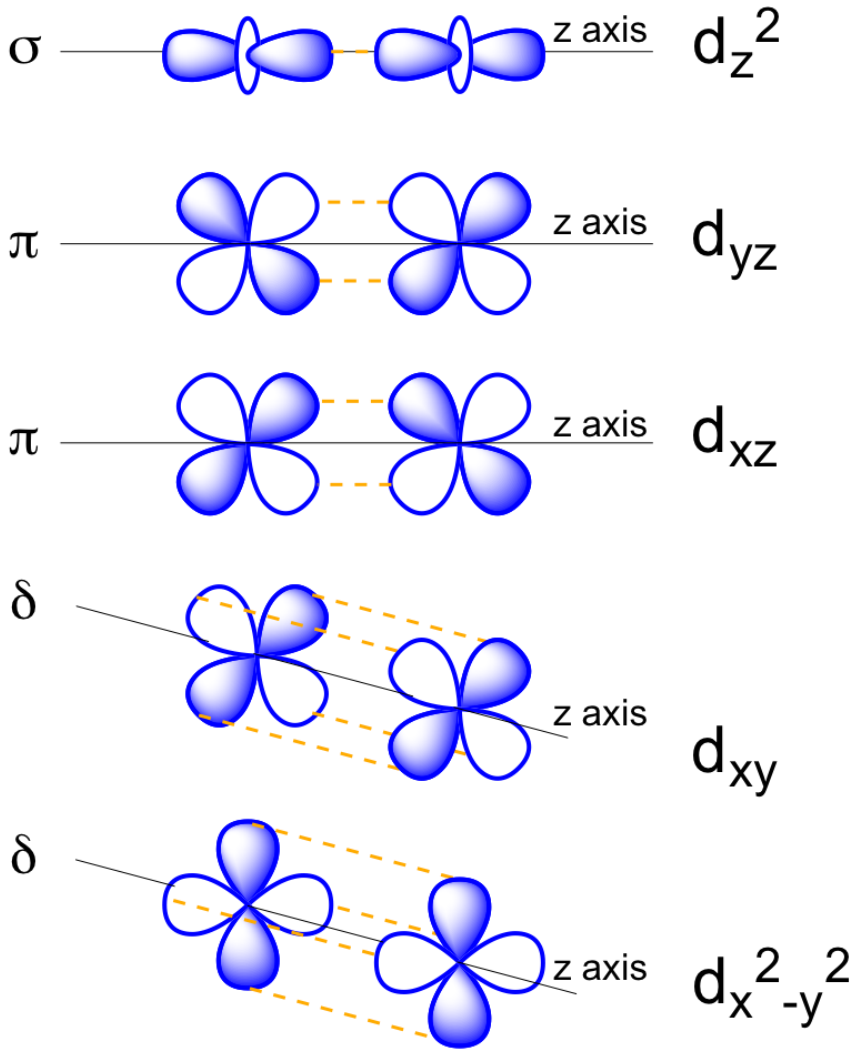
H_2O: 2σ bonds

δ bond



d ***d*** **δ bond**

Two nodal planes



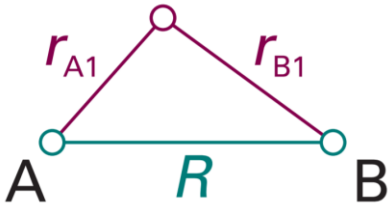
Molecular Orbital Theory

Molecular Orbitals (MO): the one-electron wavefunctions obtained by solving the Schrödinger equation $\hat{H}\psi = E\psi$.
MO gives the distribution of the electron in the molecule though the value of $|\psi|^2$.

VB theory： electrons belong to particular bonds.
MO theory: electrons spread throughout the entire molecule.

Linear combinations of atomic orbitals (LCAO)

H_2^+



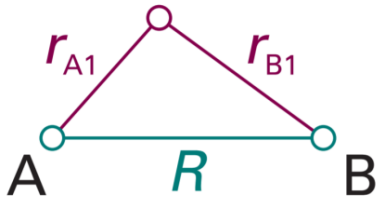
$$\psi_{\pm} = N(A \pm B)$$

A denotes χ_{H1s_A}, B denotes χ_{H1s_B}

LCAO-MO: An approximate molecular orbital formed from a linear combination of atomic orbitals

Bonding and Antibonding Orbitals

$$\psi_{\pm} = N(A \pm B)$$

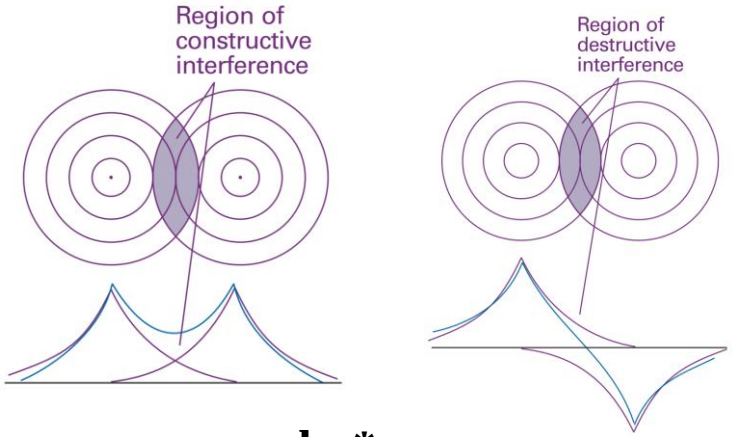


Bonding orbital: an orbital which, if occupied, helps to bind two atoms together.

$$\psi_{+}^{2} = N^{2}(A^{2} + B^{2} + 2AB)$$

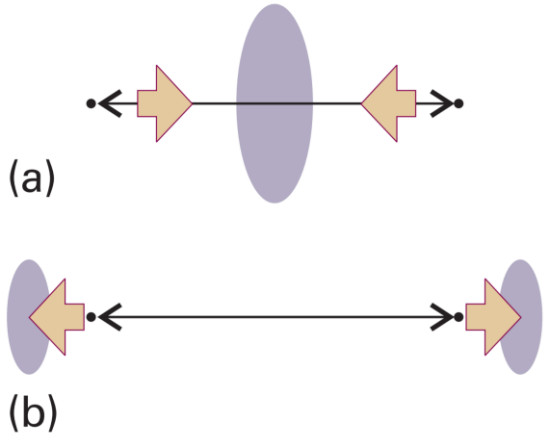
Antibonding orbital: an orbital that, if occupied, contributes to a reduction in the cohesion between two atoms and helps to **raise the energy** of the molecule relative to the separated atoms.

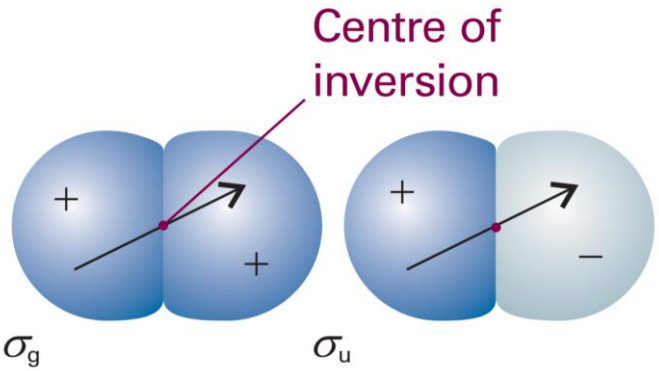
$$\psi_{-}^{2} = N^{2}(A^{2} + B^{2} - 2AB)$$



σ and σ* orbitals

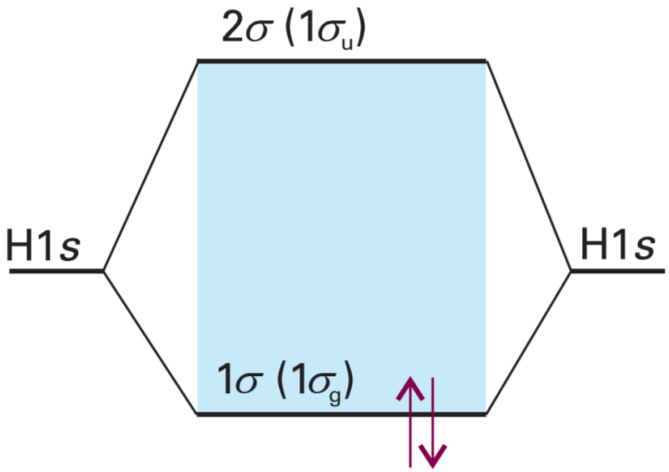
The origin of bonding and antibonding effects

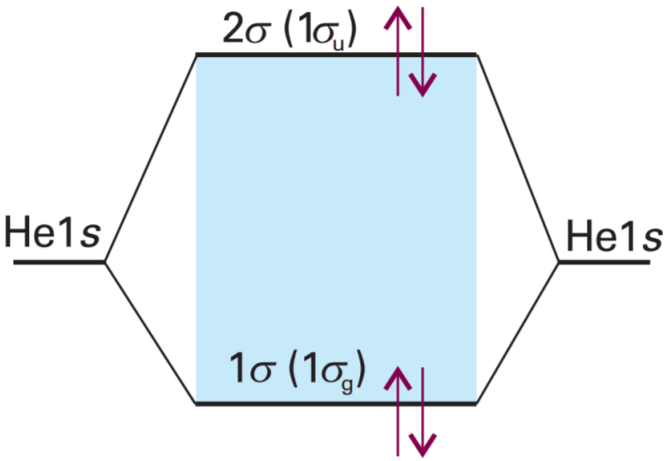




For homonuclear diatomic molecules

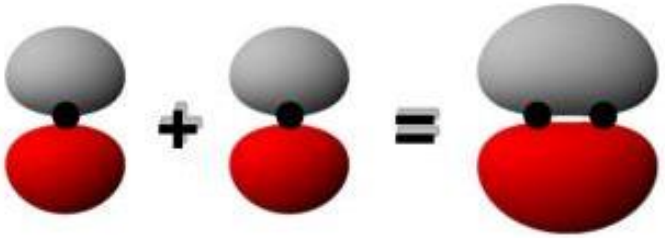
Explain why H_2 forms easily and He_2 does not?



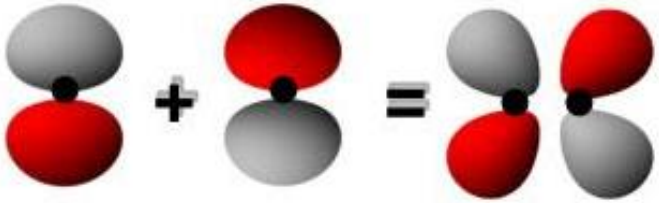


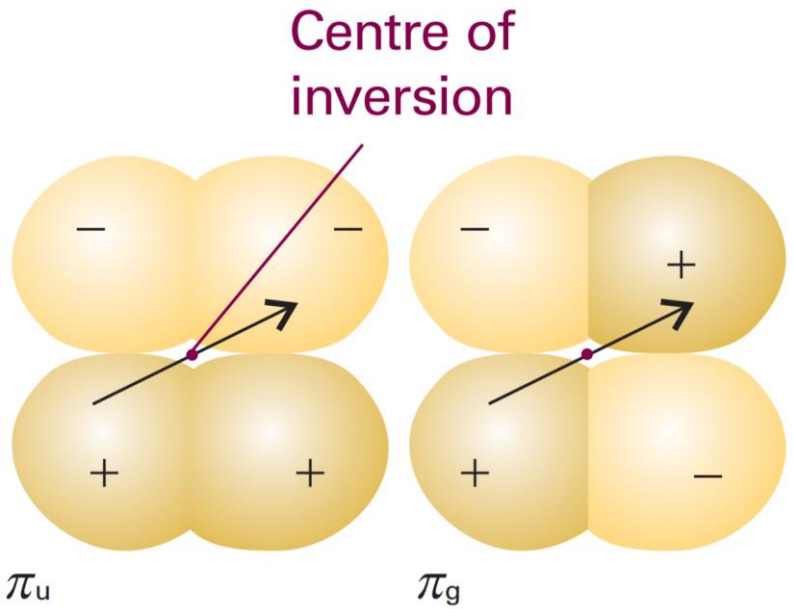
MO energy level diagrams

π bonding orbital

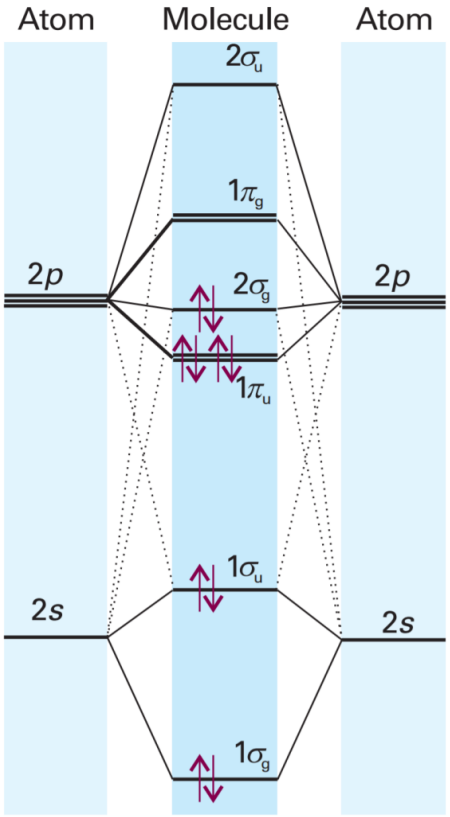


π antibonding orbital, π *

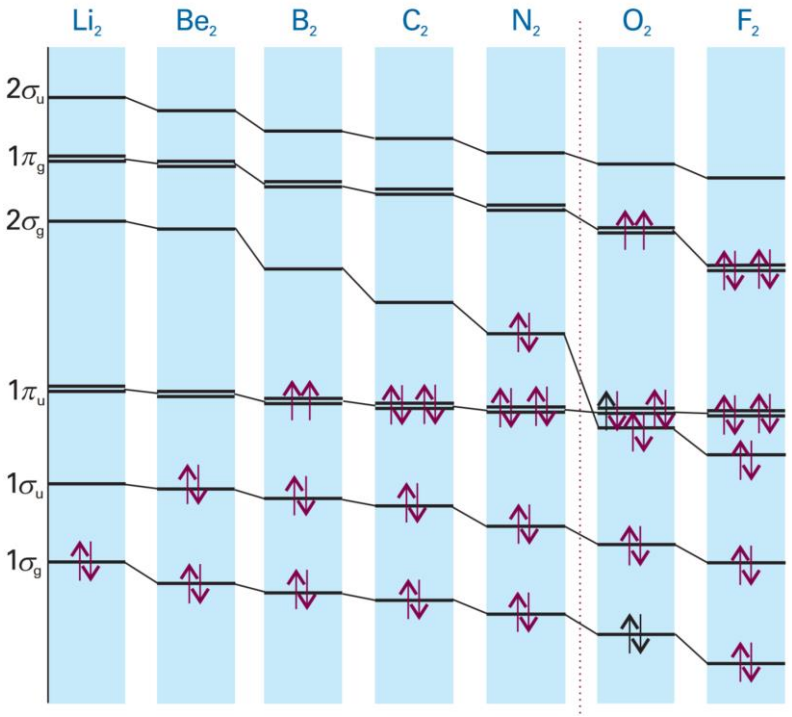


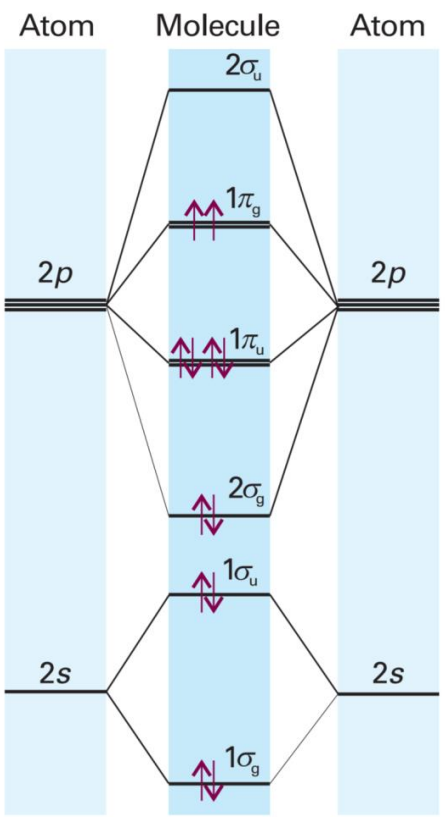


For homonuclear diatomic molecules



N_2

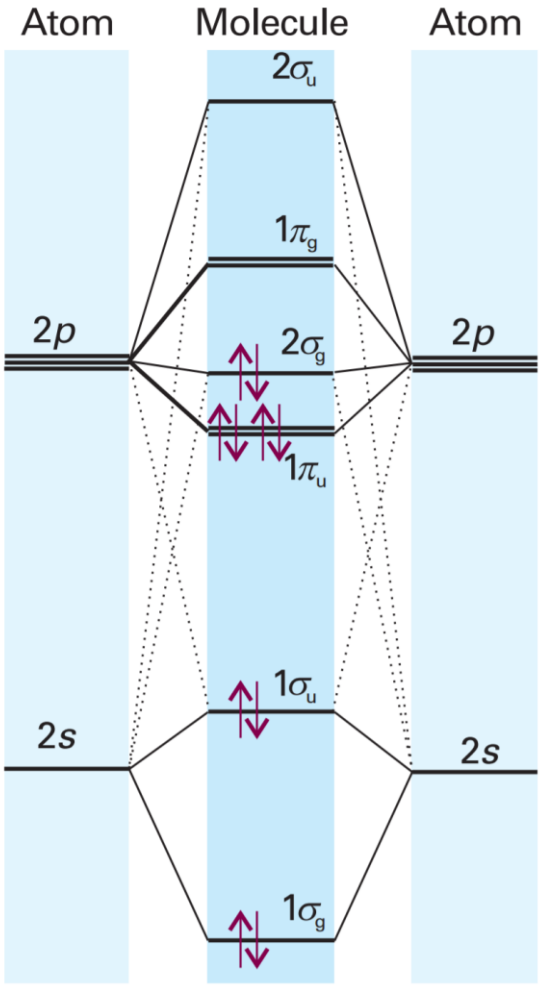




O_2

MO energy level diagrams

Electronic configuration of molecules

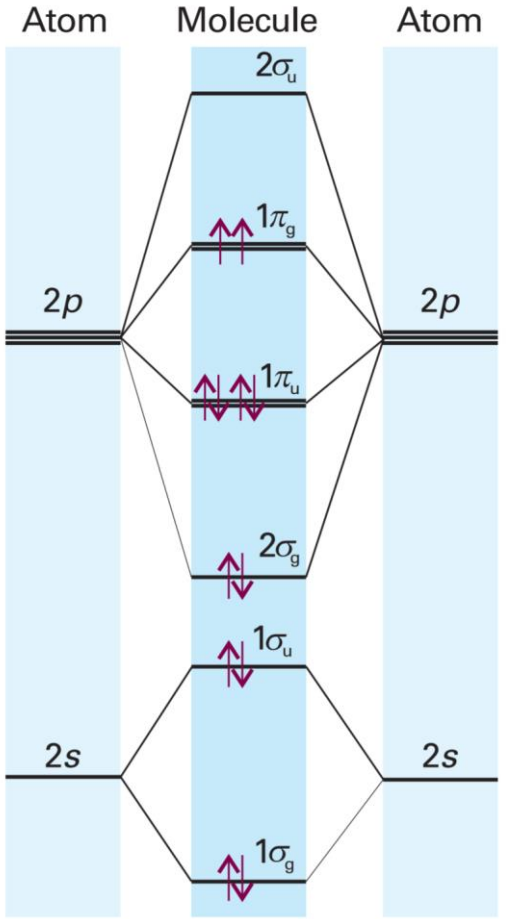


N_2: $1\sigma_g^2\ 1\sigma_u^2\ 1\pi_u^4\ 2\sigma_g^2$

O_2: $1\sigma_g^2\ 1\sigma_u^2\ 2\sigma_g^2 1\pi_u^4 1\pi_g^2$

F_2: $1\sigma_g^2\ 1\sigma_u^2\ 2\sigma_g^2 1\pi_u^4 1\pi_g^4$

Ne_2: $1\sigma_g^2\ 1\sigma_u^2\ 2\sigma_g^2 1\pi_u^4 1\pi_g^4\ 2\sigma_u^2$



The number of bonding *e* pairs shared by 2 atoms in a molecule.

Bond order:

of *e* in antibonding orbitals

$$b = \frac{1}{2}(n - n^*)$$

of *e* in bonding orbitals

For bonds between atoms of a given pair of elements:

1 The greater the bond order, the shorter the bond.
2 The greater the bond order, the greater the bond strength.

SAMPLE PROBLEM Using MO Theory to Explain Bond Properties

As the following data show, removing an electron from N_2 forms an ion with a weaker, longer bond than in the parent molecules, whereas the ion formed from O_2 has a stronger, shorter bond:

	N_2	N_2^+	O_2	O_2^+
Bond energy (kJ/mol)	945	841	498	623
Bond length (pm)	110	112	121	112

Explain these facts with diagrams that show the sequence and occupancy of MOs.

SAMPLE PROBLEM **Using MO Theory to Explain Bond Properties**

continued

